

Atmosphere Model

5.4

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Module Index

1.1 Modules

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Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

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Hierarchical Index

3.1 Class Hierarchy

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Data Structure Index

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Chapter 5

File Index

5.1 File List

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wind_velocity.cc	General base class for wind velocity models	80
wind_velocity.hh	A wind velocity model based on winds caused by rotation of the planet	81

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Chapter 6

Module Documentation

6.1 Models

Modules

- [Environment](#)

6.1.1 Detailed Description

6.2 Environment

Modules

- [Atmosphere](#)

6.2.1 Detailed Description

6.3 Atmosphere

Modules

- [BaseAtmosphere](#)

Files

- file [atmosphere_messages.hh](#)
Implement atmosphere_messages.
- file [atmosphere.hh](#)
General base class for atmosphere models.
- file [base_atmos/include/class_declarations.hh](#)
Forward declarations of classes defined for JEOD 2.0 Atmosphere.
- file [wind_velocity_base.hh](#)
General base class for wind velocity models.
- file [atmosphere_messages.cc](#)
Implement atmosphere_messages.
- file [atmosphere_state.cc](#)
Implementation of the base atmosphere-state model.
- file [wind_velocity.cc](#)
General base class for wind velocity models.
- file [wind_velocity_base.cc](#)
General base class for wind velocity models.
- file [MET/include/class_declarations.hh](#)
Forward declarations of classes defined for JEOD 2.0 Atmosphere.
- file [MET_atmosphere.hh](#)
Implement the MET atmosphere using the atmosphere framework.
- file [MET_atmosphere_state.hh](#)
Implement the MET atmosphere state using the atmosphere framework.
- file [MET_atmosphere_state_vars.hh](#)
Implement the MET atmosphere state variables using the atmosphere framework.
- file [MET_atmosphere.cc](#)
Implementation of MET atmosphere model.
- file [MET_atmosphere.cc](#)
Implementation of MET atmosphere model.
- file [MET_atmosphere_state_vars.cc](#)
Implementation of MET atmosphere model.

6.3.1 Detailed Description

6.4 BaseAtmosphere

Files

- file [atmosphere.hh](#)
General base class for atmosphere models.
- file [wind_velocity.hh](#)
A wind velocity model based on winds caused by rotation of the planet.

6.4.1 Detailed Description

Chapter 7

Namespace Documentation

7.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [Atmosphere](#)
A generic base class for atmospheres.
- class [AtmosphereMessages](#)
Describes messages used in the [Atmosphere](#) model.
- class [AtmosphereState](#)
A generic base class for atmosphere state, containing common atmosphere state parameters, i.e.
- class [WindVelocity](#)
A generic wind velocity implementation.
- class [WindVelocityBase](#)
The generic base class for wind velocity classes.
- class [WindVelocity_wind_velocity_default_data](#)
- class [METAtmosphere_solar_max_default_data](#)
- class [METAtmosphere_solar_mean_default_data](#)
- class [METAtmosphere_solar_min_default_data](#)
- class [METAtmosphereChemical](#)
The chemical composition of the MET [Atmosphere](#).
- class [METAtmosphereThermal](#)
The Thermal aspect of the computation.
- class [METAtmosphere](#)
- class [METAtmosphereState](#)
The MET specific implementation of [AtmosphereState](#).
- class [METAtmosphereStateVars](#)
The data variables component of the MET specific implementation of [AtmosphereState](#).

7.1.1 Detailed Description

Namespace jeod.

Chapter 8

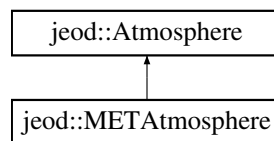
Data Structure Documentation

8.1 jeod::Atmosphere Class Reference

A generic base class for atmospheres.

```
#include <atmosphere.hh>
```

Inheritance diagram for jeod::Atmosphere:



Public Member Functions

- [Atmosphere](#) ()=default
- virtual [~Atmosphere](#) ()=default
- [Atmosphere](#) & [operator=](#) (const [Atmosphere](#) &rhs)=delete
- [Atmosphere](#) (const [Atmosphere](#) &rhs)=delete
- virtual void [update_atmosphere](#) (const PlanetFixedPosition *position, [AtmosphereState](#) *state)=0

A pure virtual function for updating the atmosphere, and inserting.

Data Fields

- bool [active](#) {true}
- If true the atmosphere state will calculate, if false it will not.*

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2p](#) (jeod, [Atmosphere](#))

8.1.1 Detailed Description

A generic base class for atmospheres.

Definition at line 78 of file atmosphere.hh.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 Atmosphere() [1/2]

```
jeod::Atmosphere::Atmosphere ( ) [default]
```

8.1.2.2 ~Atmosphere()

```
virtual jeod::Atmosphere::~~Atmosphere ( ) [virtual], [default]
```

8.1.2.3 Atmosphere() [2/2]

```
jeod::Atmosphere::Atmosphere (
    const Atmosphere & rhs ) [delete]
```

8.1.3 Member Function Documentation

8.1.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2p()

```
jeod::Atmosphere::JEOD_CLASS_ESTABLISH_FRIENDS2p (
    jeod ,
    Atmosphere ) [private]
```

8.1.3.2 operator=()

```
Atmosphere& jeod::Atmosphere::operator= (
    const Atmosphere & rhs ) [delete]
```

8.1.3.3 update_atmosphere()

```
virtual void jeod::Atmosphere::update_atmosphere (
    const PlanetFixedPosition * position,
    AtmosphereState * state ) [pure virtual]
```

A pure virtual function for updating the atmosphere, and inserting.

Parameters

in	<i>position</i>	planet fixed position
out	<i>state</i>	The AtmosphereState

Implemented in [jeod::METAtmosphere](#).

Referenced by [jeod::AtmosphereState::update_state\(\)](#).

8.1.4 Field Documentation

8.1.4.1 active

```
bool jeod::Atmosphere::active {true}
```

If true the atmosphere state will calculate, if false it will not.

trick_units(−) activity-control flag.

Definition at line 84 of file [atmosphere.hh](#).

The documentation for this class was generated from the following file:

- [atmosphere.hh](#)

8.2 jeod::AtmosphereMessages Class Reference

Describes messages used in the [Atmosphere](#) model.

```
#include <atmosphere_messages.hh>
```

Public Member Functions

- [AtmosphereMessages](#) ()=delete
- [AtmosphereMessages](#) (const [AtmosphereMessages](#) &rhs)=delete
- [AtmosphereMessages](#) & operator= (const [AtmosphereMessages](#) &rhs)=delete

Static Public Attributes

- static const char * [initialization_error](#) = "environment/atmosphere/base_atmos" "initialization_error"
Indicates an error during initialization.
- static const char * [framework_error](#) = "environment/atmosphere/base_atmos" "framework_error"
Indicates an error during use of the generic framework.
- static const char * [framework_warning](#) = "environment/atmosphere/base_atmos" "framework_warning"
Indicates a warning associated with the generic framework.
- static const char * [numerical_warning](#) = "environment/atmosphere/base_atmos" "numerical_warning"
Indicates a warning associated with numerical values.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) ([jeod](#), [AtmosphereMessages](#))

8.2.1 Detailed Description

Describes messages used in the [Atmosphere](#) model.

Definition at line 76 of file atmosphere_messages.hh.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 AtmosphereMessages() [1/2]

```
jeod::AtmosphereMessages::AtmosphereMessages ( ) [delete]
```

8.2.2.2 AtmosphereMessages() [2/2]

```
jeod::AtmosphereMessages::AtmosphereMessages (
    const AtmosphereMessages & rhs ) [delete]
```

8.2.3 Member Function Documentation

8.2.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::AtmosphereMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    AtmosphereMessages ) [private]
```

8.2.3.2 operator=()

```
AtmosphereMessages& jeod::AtmosphereMessages::operator= (
    const AtmosphereMessages & rhs ) [delete]
```

8.2.4 Field Documentation

8.2.4.1 framework_error

```
char const * jeod::AtmosphereMessages::framework_error = "environment/atmosphere/base_atmos"
"framework_error" [static]
```

Indicates an error during use of the generic framework.

trick_units(-)

Definition at line 91 of file atmosphere_messages.hh.

Referenced by jeod::WindVelocity::set_omega_scale_table(), jeod::METAtmosphere::update_atmosphere(), and jeod::WindVelocity::update_wind().

8.2.4.2 framework_warning

```
char const * jeod::AtmosphereMessages::framework_warning = "environment/atmosphere/base_atmos"
"framework_warning" [static]
```

Indicates a warning associated with the generic framework.

trick_units(-)

Definition at line 98 of file atmosphere_messages.hh.

Referenced by jeod::WindVelocityBase::update_wind().

8.2.4.3 initialization_error

```
char const * jeod::AtmosphereMessages::initialization_error = "environment/atmosphere/base_atmos"
"initialization_error" [static]
```

Indicates an error during initialization.

trick_units(-)

Definition at line 86 of file atmosphere_messages.hh.

8.2.4.4 numerical_warning

```
char const * jeod::AtmosphereMessages::numerical_warning = "environment/atmosphere/base_atmos"
"numerical_warning" [static]
```

Indicates a warning associated with numerical values.

trick_units(-)

Definition at line 103 of file atmosphere_messages.hh.

Referenced by jeod::METAtmosphere::compute_exospheric_temperature().

The documentation for this class was generated from the following files:

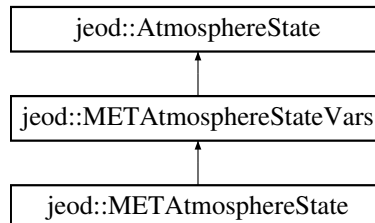
- [atmosphere_messages.hh](#)
- [atmosphere_messages.cc](#)

8.3 jeod::AtmosphereState Class Reference

A generic base class for atmosphere state, containing common atmosphere state parameters, i.e.

```
#include <atmosphere_state.hh>
```

Inheritance diagram for jeod::AtmosphereState:



Public Member Functions

- [AtmosphereState](#) ()=default
- [AtmosphereState](#) ([Atmosphere](#) &[atmos](#), const PlanetFixedPosition &[pfix_pos](#))
- virtual [~AtmosphereState](#) ()=default
- [AtmosphereState](#) & [operator=](#) (const [AtmosphereState](#) &[rhs](#))
AtmosphereState Operator =.
- [AtmosphereState](#) (const [AtmosphereState](#) &[rhs](#))
Copy Constructor.
- void [update_state](#) ([Atmosphere](#) *[atmos_model_](#), PlanetFixedPosition *[pfix_pos_](#))
Updates the invoking atmosphere state, using the atmosphere model pointed to by [atmos_model](#), and calculated at the planet fixed position pointed to by [pfix_pos](#).
- virtual void [update_state](#) ()
Updates the invoking atmosphere state, using the atmosphere model pointed to by [atmos](#), and calculated at the planet fixed position pointed to by [pfix_pos](#).
- void [update_wind](#) ([WindVelocity](#) *[wind_vel](#), double [inrtl_pos](#)[3], double [altitude](#))
Updates the wind portion of the invoking atmosphere state, using the wind model pointed to by [wind_vel](#), calculated at the inertial position given by [inrtl_pos](#) and the altitude given.

Data Fields

- bool [active](#) {true}
- double [temperature](#) {}
- double [density](#) {}
- double [pressure](#) {}
- double [wind](#) [3] {}

Protected Attributes

- [Atmosphere](#) * [atmos](#) {}
- const PlanetFixedPosition * [pfix_pos](#) {}

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [AtmosphereState](#))

8.3.1 Detailed Description

A generic base class for atmosphere state, containing common atmosphere state parameters, i.e.

pressure, density, temperature, wind velocity

Definition at line 86 of file atmosphere_state.hh.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 AtmosphereState() [1/3]

```
jeod::AtmosphereState::AtmosphereState ( ) [default]
```

8.3.2.2 AtmosphereState() [2/3]

```
jeod::AtmosphereState::AtmosphereState (
    Atmosphere & atmos,
    const PlanetFixedPosition & pfix_pos )
```

Definition at line 34 of file atmosphere_state.cc.

8.3.2.3 ~AtmosphereState()

```
virtual jeod::AtmosphereState::~~AtmosphereState ( ) [virtual], [default]
```

8.3.2.4 AtmosphereState() [3/3]

```
jeod::AtmosphereState::AtmosphereState (
    const AtmosphereState & rhs )
```

Copy Constructor.

Parameters

<i>in</i>	<i>rhs</i>	The AtmosphereState to copy from
-----------	------------	--

Definition at line 45 of file atmosphere_state.cc.

References [atmos](#), [density](#), [pfix_pos](#), [pressure](#), [temperature](#), and [wind](#).

8.3.3 Member Function Documentation**8.3.3.1 operator=()**

```
AtmosphereState & jeod::AtmosphereState::operator= (
    const AtmosphereState & rhs )
```

[AtmosphereState](#) Operator =.

Returns

The newly copied [AtmosphereState](#)

Parameters

<i>in</i>	<i>rhs</i>	The AtmosphereState to copy
-----------	------------	---

Definition at line 65 of file atmosphere_state.cc.

References [density](#), [pressure](#), and [temperature](#).

Referenced by [jeod::METAtmosphereStateVars::operator=\(\)](#).

8.3.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::AtmosphereState::p (
    jeod ,
    AtmosphereState ) [private]
```


8.3.3.3 update_state() [1/2]

```
void jeod::AtmosphereState::update_state ( ) [virtual]
```

Updates the invoking atmosphere state, using the atmosphere model pointed to by `atmos`, and calculated at the planet fixed position pointed to by `pfix_pos`.

Note that any type inheriting from [Atmosphere](#) can be used as the [Atmosphere](#) pointer but only the values associated with [AtmosphereState](#) will be copied back out.

Reimplemented in [jeod::METAtmosphereState](#).

Definition at line 107 of file `atmosphere_state.cc`.

References `active`, `atmos`, `pfix_pos`, and `jeod::Atmosphere::update_atmosphere()`.

8.3.3.4 update_state() [2/2]

```
void jeod::AtmosphereState::update_state (
    Atmosphere * atmos_model_,
    PlanetFixedPosition * pfix_pos_ )
```

Updates the invoking atmosphere state, using the atmosphere model pointed to by `atmos_model`, and calculated at the planet fixed position pointed to by `pfix_pos`.

Note that any type inheriting from [Atmosphere](#) can be sent in for `atmos_model`.

Parameters

in	<i>atmos_model_</i>	Atmosphere model.
in	<i>pfix_pos_</i>	Planetary fixed position.

Definition at line 89 of file `atmosphere_state.cc`.

References `active`, and `jeod::Atmosphere::update_atmosphere()`.

8.3.3.5 update_wind()

```
void jeod::AtmosphereState::update_wind (
    WindVelocity * wind_vel,
    double inrtl_pos[3],
    double altitude )
```

Updates the wind portion of the invoking atmosphere state, using the wind model pointed to by `wind_vel`, calculated at the inertial position given by `inrtl_pos` and the altitude given.

Parameters

in	<i>wind_vel</i>	Wind velocity model.
in	<i>inrtl_pos</i>	Current inertial position. Units: M
in	<i>altitude</i>	Geodetic (elliptic) altitude. Units: M

Definition at line 125 of file atmosphere_state.cc.

References active, jeod::WindVelocity::update_wind(), and wind.

8.3.4 Field Documentation**8.3.4.1 active**

```
bool jeod::AtmosphereState::active {true}
```

trick_units(–) Activation flag for computing state.

Definition at line 89 of file atmosphere_state.hh.

Referenced by jeod::METAtmosphereStateVars::METAtmosphereStateVars(), jeod::METAtmosphereStateVars↵
::operator=(), update_state(), jeod::METAtmosphereState::update_state(), and update_wind().

8.3.4.2 atmos

```
Atmosphere* jeod::AtmosphereState::atmos {} [protected]
```

Definition at line 100 of file atmosphere_state.hh.

Referenced by AtmosphereState(), and update_state().

8.3.4.3 density

```
double jeod::AtmosphereState::density {}
```

trick_units(kg/m3) total density at altitude

Definition at line 93 of file atmosphere_state.hh.

Referenced by jeod::METAtmosphere::atmos_MET_FAIR5(), AtmosphereState(), jeod::METAtmosphere↵
::compute_seasonal_lat_variation_He(), jeod::METAtmosphere::compute_seasonal_latitude_variation(), jeod::↵
METAtmosphere::jacchia(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.3.4.4 pfix_pos

```
const PlanetFixedPosition* jeod::AtmosphereState::pfix_pos {} [protected]
```

Definition at line 101 of file atmosphere_state.hh.

Referenced by AtmosphereState(), update_state(), and jeod::METAtmosphereState::update_state().

8.3.4.5 pressure

```
double jeod::AtmosphereState::pressure {}
```

trick_units(N/m2) Total pressure

Definition at line 95 of file atmosphere_state.hh.

Referenced by AtmosphereState(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.3.4.6 temperature

```
double jeod::AtmosphereState::temperature {}
```

trick_units(K) Temperature at altitude

Definition at line 91 of file atmosphere_state.hh.

Referenced by AtmosphereState(), jeod::METAtmosphere::jacchia(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.3.4.7 wind

```
double jeod::AtmosphereState::wind[3] {}
```

trick_units(m/s) Wind velocity

Definition at line 97 of file atmosphere_state.hh.

Referenced by AtmosphereState(), and update_wind().

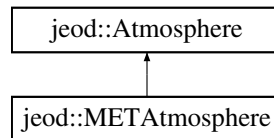
The documentation for this class was generated from the following files:

- [atmosphere_state.hh](#)
- [atmosphere_state.cc](#)

8.4 jeod::METAtmosphere Class Reference

```
#include <MET_atmosphere.hh>
```

Inheritance diagram for jeod::METAtmosphere:



Public Types

- enum [AtmosMETGeoIndexType](#) { [ATMOS_MET_GI_AP](#) = 0 , [ATMOS_MET_GI_KP](#) = 1 }

Public Member Functions

- [METAtmosphere](#) (const double &trunc_julian_time_in)
- [~METAtmosphere](#) () override=default
- [METAtmosphere](#) & operator= (const [METAtmosphere](#) &)=delete
- [METAtmosphere](#) (const [METAtmosphere](#) &)=delete
- void [update_atmosphere](#) (const PlanetFixedPosition *pfix_pos, [AtmosphereState](#) *state) override
A pure virtual function for updating the atmosphere, and inserting.
- void [update_atmosphere](#) (const PlanetFixedPosition *pfix_pos, [METAtmosphereStateVars](#) *state)
Front-end to the computation of the [METAtmosphere](#) at the current time Inserts the results into the [METAtmosphereStateVars](#) pointed to by ext_state.

Data Fields

- [AtmosMETGeoIndexType](#) geo_index_type {[ATMOS_MET_GI_AP](#)}
- double [geo_index](#) {}
- double [F10](#) {}
- double [F10B](#) {}
- [METAtmosphereChemical](#) species

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [METAtmosphere](#))
- void [update_atmosphere](#) (const PlanetFixedPosition *pfix_pos)
Calculates the [METAtmosphere](#), at the current time.
- void [modify_densities](#) ()
- void [compute_solar_angles](#) ()
- void [compute_exospheric_temperature](#) ()
- void [jacchia](#) ()
- void [compute_seasonal_latitude_variation](#) ()
- void [compute_seasonal_lat_variation_He](#) ()
- void [atmos_MET_FAIR5](#) ()
- double [compute_mol_wt](#) (double altitude)
- double [apply_gauss_quadrature](#) (int altitude_index_start, double ceiling)

Private Attributes

- double [altitude_km](#) {}
- double [latitude](#) {}
- double [longitude](#) {}
- double [barometric_equation_ceiling](#) {105.0}
- const double & [trunc_julian_time](#)
- double [tjt_year_start](#) {11544.0}
- double [fraction_of_year](#) {}
- int [day_of_year](#) {1}
- int [max_days_this_year](#) {366}
- int [year](#) {2000}
- double [solar_declination_angle](#) {}
- double [solar_hour_angle](#) {}
- [METAtmosphereStateVars](#) state
- [METAtmosphereThermal](#) thermal
- const double [fairing_k](#)

Static Private Attributes

- static constexpr double [R_gas_constant](#) {8.31432}
- static constexpr double [days_per_year](#) {365.2422}
- static constexpr double [Avogadro](#) {6.02257E23}
- static constexpr double [two_pi](#) {6.28318531}
- static constexpr double [three_pi_two](#) {4.71238898}
- static constexpr double [deg_to_rad](#) {0.017453293}
- static constexpr int [days_per_century](#) {36525}
- static constexpr int [minutes_per_day](#) {1440}
- static constexpr double [mol_weight_barometric_ceiling](#) {27.72594278125}
- static constexpr double [base_fairing_height](#) {440.0}
- static constexpr int [num_mol_wt_coeffs](#) {7}
- static const double [mol_wt_coeffs](#) [[num_mol_wt_coeffs](#)]
- static const int [num_integ_divisions](#) {8}
- static const double [gauss_altitudes](#) [[num_integ_divisions](#)+1] = {90.0, 105.0, 125.0, 160.0, 200.0, 300.0, 500.0, 1500.0, 2500.0}
- static const int [gauss_n](#) [[num_integ_divisions](#)] = {4, 5, 6, 6, 6, 6, 6}

8.4.1 Detailed Description

Definition at line 179 of file MET_atmosphere.hh.

8.4.2 Member Enumeration Documentation

8.4.2.1 AtmosMETGeoIndexType

```
enum jeod::METAtmosphere::AtmosMETGeoIndexType
```

Enumerator

ATMOS_MET_GI_AP	
ATMOS_MET_GI_KP	

Definition at line 182 of file MET_atmosphere.hh.

8.4.3 Constructor & Destructor Documentation

8.4.3.1 METAtmosphere() [1/2]

```
jeod::METAtmosphere::METAtmosphere (
    const double & trunc_julian_time_in ) [explicit]
```

Definition at line 83 of file MET_atmosphere.cc.

8.4.3.2 ~METAtmosphere()

```
jeod::METAtmosphere::~~METAtmosphere ( ) [override], [default]
```

8.4.3.3 METAtmosphere() [2/2]

```
jeod::METAtmosphere::METAtmosphere (
    const METAtmosphere & ) [delete]
```

8.4.4 Member Function Documentation

8.4.4.1 apply_gauss_quadrature()

```
double jeod::METAtmosphere::apply_gauss_quadrature (
    int altitude_index_start,
    double ceiling ) [private]
```

Definition at line 1148 of file MET_atmosphere.cc.

References [barometric_equation_ceiling](#), [compute_mol_wt\(\)](#), [jeod::METAtmosphereThermal::compute_↔ temperature\(\)](#), [gauss_altitudes](#), [gauss_n](#), and [thermal](#).

Referenced by [jacchia\(\)](#).

8.4.4.2 atmos_MET_FAIR5()

```
void jeod::METAtmosphere::atmos_MET_FAIR5 ( ) [private]
```

Definition at line 1017 of file MET_atmosphere.cc.

References altitude_km, base_fairing_height, compute_seasonal_lat_variation_He(), jeod::AtmosphereState↵::density, fairing_k, jeod::METAtmosphereChemical::num_density, species, and state.

Referenced by modify_densities().

8.4.4.3 compute_exospheric_temperature()

```
void jeod::METAtmosphere::compute_exospheric_temperature ( ) [private]
```

Definition at line 540 of file MET_atmosphere.cc.

References ATMOS_MET_GI_KP, jeod::METAtmosphereStateVars::exo_temp, F10, F10B, fraction_of_year, geo↵_index, geo_index_type, latitude, jeod::AtmosphereMessages::numerical_warning, p(), solar_declination_angle, solar_hour_angle, state, and two_pi.

Referenced by update_atmosphere().

8.4.4.4 compute_mol_wt()

```
double jeod::METAtmosphere::compute_mol_wt (
    double altitude ) [private]
```

Definition at line 1068 of file MET_atmosphere.cc.

References barometric_equation_ceiling, mol_weight_barometric_ceiling, and mol_wt_coeffs.

Referenced by apply_gauss_quadrature(), and jacchia().

8.4.4.5 compute_seasonal_lat_variation_He()

```
void jeod::METAtmosphere::compute_seasonal_lat_variation_He ( ) [private]
```

Definition at line 959 of file MET_atmosphere.cc.

References jeod::AtmosphereState::density, latitude, jeod::METAtmosphereChemical::num_density, solar_↵declination_angle, species, and state.

Referenced by atmos_MET_FAIR5(), and modify_densities().

8.4.4.6 compute_seasonal_latitude_variation()

```
void jeod::METAtmosphere::compute_seasonal_latitude_variation ( ) [private]
```

Definition at line 903 of file MET_atmosphere.cc.

References altitude_km, jeod::AtmosphereState::density, fraction_of_year, latitude, and state.

Referenced by modify_densities().

8.4.4.7 compute_solar_angles()

```
void jeod::METAtmosphere::compute_solar_angles ( ) [private]
```

Definition at line 346 of file MET_atmosphere.cc.

References day_of_year, days_per_century, days_per_year, deg_to_rad, fraction_of_year, longitude, max_days_↵
this_year, minutes_per_day, solar_declination_angle, solar_hour_angle, three_pi_two, tjt_year_start, trunc_julian_↵
_time, two_pi, and year.

Referenced by update_atmosphere().

8.4.4.8 jacchia()

```
void jeod::METAtmosphere::jacchia ( ) [private]
```

Definition at line 694 of file MET_atmosphere.cc.

References altitude_km, apply_gauss_quadrature(), Avogadro, barometric_equation_ceiling, compute_↵
mol_wt(), jeod::METAtmosphereThermal::compute_temperature(), jeod::AtmosphereState::density, jeod::↵
METAtmosphereChemical::frac, jeod::METAtmosphereChemical::mol_weight, jeod::METAtmosphereStateVars↵
::mol_weight, mol_weight_barometric_ceiling, jeod::METAtmosphereChemical::nominal_mol_weight, jeod::↵
METAtmosphereChemical::num_density, R_gas_constant, species, state, jeod::METAtmosphereThermal::T_out,↵
jeod::AtmosphereState::temperature, thermal, and jeod::METAtmosphereThermal::update().

Referenced by update_atmosphere().

8.4.4.9 modify_densities()

```
void jeod::METAtmosphere::modify_densities ( ) [private]
```

Definition at line 307 of file MET_atmosphere.cc.

References altitude_km, atmos_MET_FAIR5(), base_fairing_height, compute_seasonal_lat_variation_He(), and
compute_seasonal_latitude_variation().

Referenced by update_atmosphere().

8.4.4.10 operator=()

```
METAtmosphere& jeod::METAtmosphere::operator= (
    const METAtmosphere & ) [delete]
```

8.4.4.11 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::METAtmosphere::p (
    jeod ,
    METAtmosphere ) [private]
```

Referenced by `compute_exospheric_temperature()`.

8.4.4.12 update_atmosphere() [1/3]

```
void jeod::METAtmosphere::update_atmosphere (
    const PlanetFixedPosition * pfix_pos ) [private]
```

Calculates the [METAtmosphere](#), at the current time.

Parameters

in	<i>pfix_pos</i>	Geodetic altitude, latitude and longitude.
----	-----------------	--

Definition at line 261 of file `MET_atmosphere.cc`.

References `jeod::METAtmosphereStateVars::A`, `altitude_km`, `compute_exospheric_temperature()`, `compute_solar_angles()`, `jeod::AtmosphereState::density`, `jeod::AtmosphereMessages::framework_error`, `jeod::METAtmosphereStateVars::He`, `jeod::METAtmosphereStateVars::Hyd`, `jacchia()`, `latitude`, `jeod::METAtmosphereStateVars::log10_dens`, `longitude`, `modify_densities()`, `jeod::METAtmosphereStateVars::mol_weight`, `jeod::METAtmosphereStateVars::N2`, `jeod::METAtmosphereChemical::num_density`, `jeod::METAtmosphereStateVars::Ox`, `jeod::METAtmosphereStateVars::Ox2`, `jeod::AtmosphereState::pressure`, `R_gas_constant`, `species`, `state`, and `jeod::AtmosphereState::temperature`.

8.4.4.13 update_atmosphere() [2/3]

```
void jeod::METAtmosphere::update_atmosphere (
    const PlanetFixedPosition * position,
    AtmosphereState * state ) [override], [virtual]
```

A pure virtual function for updating the atmosphere, and inserting.

Parameters

in	<i>position</i>	planet fixed position
out	<i>state</i>	The AtmosphereState

Implements [jeod::Atmosphere](#).

Definition at line 204 of file MET_atmosphere.cc.

References [jeod::AtmosphereMessages::framework_error](#), and [state](#).

Referenced by [update_atmosphere\(\)](#), and [jeod::METAtmosphereState::update_state\(\)](#).

8.4.4.14 update_atmosphere() [3/3]

```
void jeod::METAtmosphere::update_atmosphere (
    const PlanetFixedPosition * pfix_pos,
    METAtmosphereStateVars * ext_state )
```

Front-end to the computation of the [METAtmosphere](#) at the current time Inserts the results into the [METAtmosphereStateVars](#) pointed to by *ext_state*.

This function is for a [METAtmosphereStateVars](#).

Parameters

in	<i>pfix_pos</i>	Geodetic altitude, latitude and longitude.
out	<i>ext_state</i>	Where the state results will be sent.

Definition at line 239 of file MET_atmosphere.cc.

References [jeod::AtmosphereMessages::framework_error](#), [state](#), and [update_atmosphere\(\)](#).

8.4.5 Field Documentation

8.4.5.1 altitude_km

```
double jeod::METAtmosphere::altitude_km {} [private]
```

trick_units(km) Copy of vehicle altitude

Definition at line 202 of file MET_atmosphere.hh.

Referenced by [atmos_MET_FAIR5\(\)](#), [compute_seasonal_latitude_variation\(\)](#), [jacchia\(\)](#), [modify_densities\(\)](#), and [update_atmosphere\(\)](#).

8.4.5.2 Avogadro

```
constexpr double jeod::METAtmosphere::Avogadro {6.02257E23} [static], [constexpr], [private]
```

trick_units(–) Avogadros number

Definition at line 243 of file MET_atmosphere.hh.

Referenced by jacchia().

8.4.5.3 barometric_equation_ceiling

```
double jeod::METAtmosphere::barometric_equation_ceiling {105.0} [private]
```

trick_units(km) the ceiling for integration using the barometric equation. Above this value, the integration switches to the diffusion equation. Value is 105km in the 1970 paper and 100km in the 1971 paper.

Definition at line 206 of file MET_atmosphere.hh.

Referenced by apply_gauss_quadrature(), compute_mol_wt(), and jacchia().

8.4.5.4 base_fairing_height

```
constexpr double jeod::METAtmosphere::base_fairing_height {440.0} [static], [constexpr],  
[private]
```

trick_units(km) Altitude at which to start fairing between the lower altitude which has no seasonal-latitude Helium density variation, and the upper atmosphere – starting at 500km – which does.

Definition at line 258 of file MET_atmosphere.hh.

Referenced by atmos_MET_FAIR5(), and modify_densities().

8.4.5.5 day_of_year

```
int jeod::METAtmosphere::day_of_year {1} [private]
```

trick_units(count) day number since start of year.

Definition at line 221 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.6 days_per_century

```
constexpr int jeod::METAtmosphere::days_per_century {36525} [static], [constexpr], [private]
```

trick_units(count) days per century

Definition at line 250 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.7 days_per_year

```
constexpr double jeod::METAtmosphere::days_per_year {365.2422} [static], [constexpr], [private]
```

trick_units(day) days per year

Definition at line 242 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.8 deg_to_rad

```
constexpr double jeod::METAtmosphere::deg_to_rad {0.017453293} [static], [constexpr], [private]
```

trick_units(degree/rad) degree-to-radian conversion

Definition at line 249 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.9 F10

```
double jeod::METAtmosphere::F10 {}
```

trick_units(-) Solar radio noise flux.

Definition at line 195 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), jeod::METAtmosphere_solar_max_default_data::initialize(), jeod::METAtmosphere_solar_mean_default_data::initialize(), and jeod::METAtmosphere_solar_min_default_data::initialize().

8.4.5.10 F10B

```
double jeod::METAtmosphere::F10B {}
```

trick_units(–) 90 day average of solar radio noise flux.

Definition at line 197 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), jeod::METAtmosphere_solar_max_default_data::initialize(), jeod::METAtmosphere_solar_mean_default_data::initialize(), and jeod::METAtmosphere_solar_min_default_data::initialize().

8.4.5.11 fairing_k

```
const double jeod::METAtmosphere::fairing_k [private]
```

trick_units(rad/km) Factor which, when multiplied by the altitude delta above the base-fairing-height provides an angle. The square of the cosine of that angle indicates how much of the seasonal-variation in Helium density to apply. $\text{density} = \text{corrected-density} * (\text{non-corrected-density} / \text{corrected-density}) ^ (\cos^2 (\text{fairing_k} * \text{delta-altitude}))$ At base-fairing-height, none gets applied. By 500km, it all gets applied.

Definition at line 262 of file MET_atmosphere.hh.

Referenced by atmos_MET_FAIR5().

8.4.5.12 fraction_of_year

```
double jeod::METAtmosphere::fraction_of_year {} [private]
```

trick_units(–) fraction of this year that has passed.

Definition at line 219 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), compute_seasonal_latitude_variation(), and compute_solar_angles().

8.4.5.13 gauss_altitudes

```
const double jeod::METAtmosphere::gauss_altitudes = {90.0, 105.0, 125.0, 160.0, 200.0, 300.0, 500.0, 1500.0, 2500.0} [static], [private]
```

trick_units(–) The boundaries of the cells that are used to break down the integration over the atmosphere into more manageable pieces. NOTE - gauss_altitudes[1] must mark the upper limit of the altitude over which the barometric equation is valid, this is either 100km or 105km, depending on which paper is used; gauss_altitude[6] must be equal to 500km.

Definition at line 283 of file MET_atmosphere.hh.

Referenced by apply_gauss_quadrature().

8.4.5.14 gauss_n

```
const int jeod::METAtmosphere::gauss_n = {4, 5, 6, 6, 6, 6, 6, 6} [static], [private]
```

trick_units(–) The number of data-points to be used for the gauss-quadrature integration for each interval defined in the gauss_altitudes array. AKA the order of the gauss-quadrature.

Definition at line 290 of file MET_atmosphere.hh.

Referenced by apply_gauss_quadrature().

8.4.5.15 geo_index

```
double jeod::METAtmosphere::geo_index {}
```

trick_units(–) Geomagnetic variations index (Ap or Kp).

Definition at line 193 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), jeod::METAtmosphere_solar_max_default_data::initialize(), jeod::METAtmosphere_solar_mean_default_data::initialize(), and jeod::METAtmosphere_solar_min_default_data::initialize().

8.4.5.16 geo_index_type

```
AtmosMETGeoIndexType jeod::METAtmosphere::geo_index_type {ATMOS_MET_GI_AP}
```

Definition at line 190 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), jeod::METAtmosphere_solar_max_default_data::initialize(), jeod::METAtmosphere_solar_mean_default_data::initialize(), and jeod::METAtmosphere_solar_min_default_data::initialize().

8.4.5.17 latitude

```
double jeod::METAtmosphere::latitude {} [private]
```

trick_units(rad) Copy of vehicle latitude

Definition at line 203 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), compute_seasonal_lat_variation_He(), compute_seasonal_latitude_variation(), and update_atmosphere().

8.4.5.18 longitude

```
double jeod::METAtmosphere::longitude {} [private]
```

trick_units(rad) Copy of vehicle longitude

Definition at line 204 of file MET_atmosphere.hh.

Referenced by compute_solar_angles(), and update_atmosphere().

8.4.5.19 max_days_this_year

```
int jeod::METAtmosphere::max_days_this_year {366} [private]
```

trick_units(count) number of days this year (365 or 366)

Definition at line 223 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.20 minutes_per_day

```
constexpr int jeod::METAtmosphere::minutes_per_day {1440} [static], [constexpr], [private]
```

trick_units(count) minutes per day

Definition at line 251 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.21 mol_weight_barometric_ceiling

```
constexpr double jeod::METAtmosphere::mol_weight_barometric_ceiling {27.72594278125} [static],  
[constexpr], [private]
```

trick_units(g/mol) mean molar mass at barometric-ceiling and higher.

Definition at line 254 of file MET_atmosphere.hh.

Referenced by compute_mol_wt(), and jacchia().

8.4.5.22 mol_wt_coeffs

```
const double jeod::METAtmosphere::mol_wt_coeffs [static], [private]
```

Initial value:

```
= {28.15204, -0.085586, 1.284E-4, -1.0056E-5, -1.021E-5, 1.5044E-6, 9.9826E-8}
```

trick_units(–) polynomial coefficients for computing the molecular weights in the region where the barometric equation is used.

Definition at line 275 of file MET_atmosphere.hh.

Referenced by compute_mol_wt().

8.4.5.23 num_integ_divisions

```
const int jeod::METAtmosphere::num_integ_divisions {8} [static], [private]
```

trick_units(count) the number of altitude bins used for dividing the atmosphere into manageable pieces.

Definition at line 280 of file MET_atmosphere.hh.

8.4.5.24 num_mol_wt_coeffs

```
constexpr int jeod::METAtmosphere::num_mol_wt_coeffs {7} [static], [constexpr], [private]
```

trick_units(count) the number of polynomial coefficients.

Definition at line 274 of file MET_atmosphere.hh.

8.4.5.25 R_gas_constant

```
constexpr double jeod::METAtmosphere::R_gas_constant {8.31432} [static], [constexpr], [private]
```

trick_units(J/(mol*K)) R

Definition at line 238 of file MET_atmosphere.hh.

Referenced by jacchia(), and update_atmosphere().

8.4.5.26 solar_declination_angle

```
double jeod::METAtmosphere::solar_declination_angle {} [private]
```

trick_units(rad) declination angle

Definition at line 227 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), compute_seasonal_lat_variation_He(), and compute_solar_angles().

8.4.5.27 solar_hour_angle

```
double jeod::METAtmosphere::solar_hour_angle {} [private]
```

trick_units(rad) solar hour angle

Definition at line 229 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), and compute_solar_angles().

8.4.5.28 species

```
METAtmosphereChemical jeod::METAtmosphere::species
```

trick_units(-) The chemical composition of the atmosphere.

Definition at line 199 of file MET_atmosphere.hh.

Referenced by atmos_MET_FAIR5(), compute_seasonal_lat_variation_He(), jacchia(), and update_atmosphere().

8.4.5.29 state

```
METAtmosphereStateVars jeod::METAtmosphere::state [private]
```

trick_units(-) A scratch set of state variables, used for populating state variables internally before being copied onto the real state.

Definition at line 231 of file MET_atmosphere.hh.

Referenced by atmos_MET_FAIR5(), compute_exospheric_temperature(), compute_seasonal_lat_variation_He(), compute_seasonal_latitude_variation(), jacchia(), and update_atmosphere().

8.4.5.30 thermal

```
METAtmosphereThermal jeod::METAtmosphere::thermal [private]
```

trick_units(-) Thermal aspect of the model

Definition at line 235 of file MET_atmosphere.hh.

Referenced by apply_gauss_quadrature(), and jacchia().

8.4.5.31 three_pi_two

```
constexpr double jeod::METAtmosphere::three_pi_two {4.71238898} [static], [constexpr], [private]
```

trick_units(-) 1.5 pi

Definition at line 248 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.32 tjt_year_start

```
double jeod::METAtmosphere::tjt_year_start {11544.0} [private]
```

trick_units(day) value of trunc_julian_time at the start of the current year.

Definition at line 215 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.33 trunc_julian_time

```
const double& jeod::METAtmosphere::trunc_julian_time [private]
```

trick_units(day) Current time

Definition at line 214 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

8.4.5.34 two_pi

```
constexpr double jeod::METAtmosphere::two_pi {6.28318531} [static], [constexpr], [private]
```

trick_units(-) 2 pi

Definition at line 247 of file MET_atmosphere.hh.

Referenced by compute_exospheric_temperature(), and compute_solar_angles().

8.4.5.35 year

```
int jeod::METAtmosphere::year {2000} [private]
```

trick_units(count) current year identifier

Definition at line 225 of file MET_atmosphere.hh.

Referenced by compute_solar_angles().

The documentation for this class was generated from the following files:

- [MET_atmosphere.hh](#)
- [MET_atmosphere.cc](#)

8.5 jeod::METAtmosphere_solar_max_default_data Class Reference

```
#include <solar_max.hh>
```

Public Member Functions

- void [initialize](#) (METAtmosphere *)

8.5.1 Detailed Description

Definition at line 55 of file solar_max.hh.

8.5.2 Member Function Documentation

8.5.2.1 initialize()

```
void jeod::METAtmosphere_solar_max_default_data::initialize (
    METAtmosphere * METAtmosphere_ptr )
```

Definition at line 35 of file solar_max.cc.

References [jeod::METAtmosphere::ATMOS_MET_GI_AP](#), [jeod::METAtmosphere::F10](#), [jeod::METAtmosphere::F10B](#), [jeod::METAtmosphere::geo_index](#), and [jeod::METAtmosphere::geo_index_type](#).

The documentation for this class was generated from the following files:

- [solar_max.hh](#)
- [solar_max.cc](#)

8.6 jeod::METAtmosphere_solar_mean_default_data Class Reference

```
#include <solar_mean.hh>
```

Public Member Functions

- void [initialize](#) ([METAtmosphere](#) *)

8.6.1 Detailed Description

Definition at line 55 of file solar_mean.hh.

8.6.2 Member Function Documentation

8.6.2.1 initialize()

```
void jeod::METAtmosphere_solar_mean_default_data::initialize (
    METAtmosphere * METAtmosphere_ptr )
```

Definition at line 35 of file solar_mean.cc.

References [jeod::METAtmosphere::ATMOS_MET_GI_AP](#), [jeod::METAtmosphere::F10](#), [jeod::METAtmosphere::F10B](#), [jeod::METAtmosphere::geo_index](#), and [jeod::METAtmosphere::geo_index_type](#).

The documentation for this class was generated from the following files:

- [solar_mean.hh](#)
- [solar_mean.cc](#)

8.7 jeod::METAtmosphere_solar_min_default_data Class Reference

```
#include <solar_min.hh>
```

Public Member Functions

- void [initialize](#) ([METAtmosphere](#) *)

8.7.1 Detailed Description

Definition at line 55 of file [solar_min.hh](#).

8.7.2 Member Function Documentation

8.7.2.1 initialize()

```
void jeod::METAtmosphere_solar_min_default_data::initialize (  
    METAtmosphere * METAtmosphere_ptr )
```

Definition at line 35 of file [solar_min.cc](#).

References [jeod::METAtmosphere::ATMOS_MET_GI_AP](#), [jeod::METAtmosphere::F10](#), [jeod::METAtmosphere::F10B](#), [jeod::METAtmosphere::geo_index](#), and [jeod::METAtmosphere::geo_index_type](#).

The documentation for this class was generated from the following files:

- [solar_min.hh](#)
- [solar_min.cc](#)

8.8 jeod::METAtmosphereChemical Class Reference

The chemical composition of the MET [Atmosphere](#).

```
#include <MET_atmosphere.hh>
```

Public Member Functions

- [METAtmosphereChemical](#) ()=default
- virtual [~METAtmosphereChemical](#) ()=default
- [METAtmosphereChemical](#) & [operator=](#) (const [METAtmosphereChemical](#) &)=delete
- [METAtmosphereChemical](#) (const [METAtmosphereChemical](#) &)=delete

Data Fields

- double `num_density` [`num_species`] {}
- double `frac` [`num_species`]
- double `mol_weight` [`num_species`]

Static Public Attributes

- static constexpr int `num_species` {6}
- static constexpr double `nominal_mol_weight` {28.96}

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2` `p` (`jeod`, `METAtmosphereChemical`)

8.8.1 Detailed Description

The chemical composition of the MET `Atmosphere`.

Definition at line 87 of file `MET_atmosphere.hh`.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 `METAtmosphereChemical()` [1/2]

```
jeod::METAtmosphereChemical::METAtmosphereChemical ( ) [default]
```

8.8.2.2 `~METAtmosphereChemical()`

```
virtual jeod::METAtmosphereChemical::~~METAtmosphereChemical ( ) [virtual], [default]
```

8.8.2.3 `METAtmosphereChemical()` [2/2]

```
jeod::METAtmosphereChemical::METAtmosphereChemical (
    const METAtmosphereChemical & ) [delete]
```

8.8.3 Member Function Documentation

8.8.3.1 operator=()

```
METAtmosphereChemical& jeod::METAtmosphereChemical::operator= (
    const METAtmosphereChemical & ) [delete]
```

8.8.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::METAtmosphereChemical::p (
    jeod ,
    METAtmosphereChemical ) [private]
```

8.8.4 Field Documentation

8.8.4.1 frac

```
double jeod::METAtmosphereChemical::frac[num_species]
```

Initial value:

```
{
    0.78110,
    0.20955,
    0.0,
    0.0093432,
    1.289E-05,

    0.0
}
```

Definition at line 95 of file MET_atmosphere.hh.

Referenced by jeod::METAtmosphere::jacchia().

8.8.4.2 mol_weight

```
double jeod::METAtmosphereChemical::mol_weight[num_species]
```

Initial value:

```
{
    28.0134,
    31.9988,
    15.9994,
    39.948,
    4.0026,
    1.00797
}
```

Definition at line 106 of file MET_atmosphere.hh.

Referenced by jeod::METAtmosphere::jacchia().

8.8.4.3 nominal_mol_weight

```
constexpr double jeod::METAtmosphereChemical::nominal_mol_weight {28.96} [static], [constexpr]
```

Definition at line 115 of file MET_atmosphere.hh.

Referenced by jeod::METAtmosphere::jacchia().

8.8.4.4 num_density

```
double jeod::METAtmosphereChemical::num_density[num_species] {}
```

Definition at line 92 of file MET_atmosphere.hh.

Referenced by jeod::METAtmosphere::atmos_MET_FAIR5(), jeod::METAtmosphere::compute_seasonal_lat_variation_He(), jeod::METAtmosphere::jacchia(), and jeod::METAtmosphere::update_atmosphere().

8.8.4.5 num_species

```
constexpr int jeod::METAtmosphereChemical::num_species {6} [static], [constexpr]
```

Definition at line 90 of file MET_atmosphere.hh.

The documentation for this class was generated from the following file:

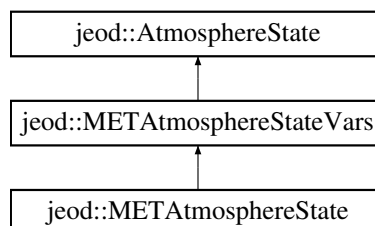
- [MET_atmosphere.hh](#)

8.9 jeod::METAtmosphereState Class Reference

The MET specific implementation of [AtmosphereState](#).

```
#include <MET_atmosphere_state.hh>
```

Inheritance diagram for jeod::METAtmosphereState:



Public Member Functions

- [METAtmosphereState](#) ([METAtmosphere](#) &atmos_model, const PlanetFixedPosition &pfix_pos)
- [METAtmosphereState](#) ()=default
- [~METAtmosphereState](#) () override=default
- [METAtmosphereState](#) & operator= (const [METAtmosphereState](#) &)=delete
- [METAtmosphereState](#) (const [METAtmosphereState](#) &)=delete
- void [update_state](#) ([METAtmosphere](#) *atmos_model, const PlanetFixedPosition *pfix_pos)
Updates the [METAtmosphereState](#) from the [METAtmosphere](#) pointed to by atmos_model_.
- void [update_state](#) () override
Updates the [METAtmosphereState](#) from the [METAtmosphere](#) pointed to by class member atmos_model using class member pointer pfix_pos.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [METAtmosphereState](#))

Private Attributes

- [METAtmosphere](#) * met_atmos {}

Additional Inherited Members

8.9.1 Detailed Description

The MET specific implementation of [AtmosphereState](#).

Definition at line 84 of file MET_atmosphere_state.hh.

8.9.2 Constructor & Destructor Documentation

8.9.2.1 METAtmosphereState() [1/3]

```
jeod::METAtmosphereState::METAtmosphereState (
    METAtmosphere & atmos_model,
    const PlanetFixedPosition & pfix_pos )
```

Definition at line 52 of file MET_atmosphere_state.cc.

8.9.2.2 METAtmosphereState() [2/3]

```
jeod::METAtmosphereState::METAtmosphereState ( ) [default]
```

8.9.2.3 ~METAtmosphereState()

```
jeod::METAtmosphereState::~~METAtmosphereState ( ) [override], [default]
```

8.9.2.4 METAtmosphereState() [3/3]

```
jeod::METAtmosphereState::METAtmosphereState (
    const METAtmosphereState & ) [delete]
```

8.9.3 Member Function Documentation

8.9.3.1 operator=()

```
METAtmosphereState& jeod::METAtmosphereState::operator= (
    const METAtmosphereState & ) [delete]
```

8.9.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::METAtmosphereState::p (
    jeod ,
    METAtmosphereState ) [private]
```

8.9.3.3 update_state() [1/2]

```
void jeod::METAtmosphereState::update_state ( ) [override], [virtual]
```

Updates the [METAtmosphereState](#) from the [METAtmosphere](#) pointed to by class member `atmos_model` using class member pointer `pfix_pos`.

This is a specific function for the case of an [METAtmosphere](#) state updating an [METAtmosphere](#) when constructed with the pointers set.

Reimplemented from [jeod::AtmosphereState](#).

Definition at line 81 of file `MET_atmosphere_state.cc`.

References [jeod::AtmosphereState::active](#), [met_atmos](#), [jeod::AtmosphereState::pfix_pos](#), and [jeod::METAtmosphereState::update_atmosphere\(\)](#).

8.9.3.4 update_state() [2/2]

```
void jeod::METAtmosphereState::update_state (
    METAtmosphere * atmos_model_,
    const PlanetFixedPosition * pfix_pos_ )
```

Updates the [METAtmosphereState](#) from the [METAtmosphere](#) pointed to by `atmos_model_`.

This is a specific function for the case of an [METAtmosphere](#) state updating an [METAtmosphere](#)

Parameters

in	<i>atmos_↔ model_</i>	METAtmosphere Model.
in	<i>pfix_pos_</i>	Current vehicle position.

Definition at line 66 of file MET_atmosphere_state.cc.

References [jeod::AtmosphereState::active](#), and [jeod::METAtmosphere::update_atmosphere\(\)](#).

8.9.4 Field Documentation

8.9.4.1 met_atmos

```
METAtmosphere* jeod::METAtmosphereState::met_atmos {} [private]
```

Definition at line 87 of file MET_atmosphere_state.hh.

Referenced by [update_state\(\)](#).

The documentation for this class was generated from the following files:

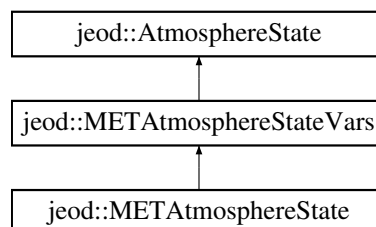
- [MET_atmosphere_state.hh](#)
- [MET_atmosphere_state.cc](#)

8.10 jeod::METAtmosphereStateVars Class Reference

The data variables component of the MET specific implementation of [AtmosphereState](#).

```
#include <MET_atmosphere_state_vars.hh>
```

Inheritance diagram for [jeod::METAtmosphereStateVars](#):



Public Member Functions

- [METAtmosphereStateVars](#) ()=default
- [METAtmosphereStateVars](#) ([Atmosphere](#) &atmos_model, const PlanetFixedPosition &pfix_pos)
- [~METAtmosphereStateVars](#) () override=default
- [METAtmosphereStateVars](#) (const [METAtmosphereStateVars](#) &rhs)
Copy Constructor.
- [METAtmosphereStateVars](#) & operator= (const [METAtmosphereStateVars](#) &rhs)
[METAtmosphereStateVars](#) operator =.

Data Fields

- double [exo_temp](#) {}
- double [log10_dens](#) {}
- double [mol_weight](#) {}
- double [N2](#) {}
- double [Ox2](#) {}
- double [Ox](#) {}
- double [A](#) {}
- double [He](#) {}
- double [Hyd](#) {}

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [METAtmosphereStateVars](#))

Additional Inherited Members

8.10.1 Detailed Description

The data variables component of the MET specific implementation of [AtmosphereState](#).

Definition at line 82 of file MET_atmosphere_state_vars.hh.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 METAtmosphereStateVars() [1/3]

```
jeod::METAtmosphereStateVars::METAtmosphereStateVars ( ) [default]
```

8.10.2.2 METAtmosphereStateVars() [2/3]

```
jeod::METAtmosphereStateVars::METAtmosphereStateVars (
    Atmosphere & atmos_model,
    const PlanetFixedPosition & pfix_pos )
```

Definition at line 49 of file MET_atmosphere_state_vars.cc.

8.10.2.3 ~METAtmosphereStateVars()

```
jeod::METAtmosphereStateVars::~~METAtmosphereStateVars ( ) [override], [default]
```

8.10.2.4 METAtmosphereStateVars() [3/3]

```
jeod::METAtmosphereStateVars::METAtmosphereStateVars (
    const METAtmosphereStateVars & rhs )
```

Copy Constructor.

Parameters

in	rhs	The METAtmosphereStateVars to copy
----	-----	--

Definition at line 59 of file MET_atmosphere_state_vars.cc.

References [A](#), [jeod::AtmosphereState::active](#), [exo_temp](#), [He](#), [Hyd](#), [log10_dens](#), [mol_weight](#), [N2](#), [Ox](#), and [Ox2](#).

8.10.3 Member Function Documentation

8.10.3.1 operator=()

```
METAtmosphereStateVars & jeod::METAtmosphereStateVars::operator= (
    const METAtmosphereStateVars & rhs )
```

[METAtmosphereStateVars](#) operator =.

Returns

The newly copied into [METAtmosphereStateVars](#)

Parameters

<i>in</i>	<i>rhs</i>	The METAtmosphereStateVars to copy from
-----------	------------	---

Definition at line 80 of file MET_atmosphere_state_vars.cc.

References [A](#), [jeod::AtmosphereState::active](#), [exo_temp](#), [He](#), [Hyd](#), [log10_dens](#), [mol_weight](#), [N2](#), [jeod::AtmosphereState::operator=\(\)](#), [Ox](#), and [Ox2](#).

8.10.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::METAtmosphereStateVars::p (
    jeod ,
    METAtmosphereStateVars ) [private]
```

8.10.4 Field Documentation**8.10.4.1 A**

```
double jeod::METAtmosphereStateVars::A {}
```

trick_units(–) A number density

Definition at line 91 of file MET_atmosphere_state_vars.hh.

Referenced by [METAtmosphereStateVars\(\)](#), [operator=\(\)](#), and [jeod::METAtmosphere::update_atmosphere\(\)](#).

8.10.4.2 exo_temp

```
double jeod::METAtmosphereStateVars::exo_temp {}
```

trick_units(K) Exospheric temperature

Definition at line 85 of file MET_atmosphere_state_vars.hh.

Referenced by [jeod::METAtmosphere::compute_exospheric_temperature\(\)](#), [METAtmosphereStateVars\(\)](#), and [operator=\(\)](#).

8.10.4.3 He

```
double jeod::METAtmosphereStateVars::He {}
```

trick_units(–) He number density

Definition at line 92 of file MET_atmosphere_state_vars.hh.

Referenced by METAtmosphereStateVars(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.10.4.4 Hyd

```
double jeod::METAtmosphereStateVars::Hyd {}
```

trick_units(–) H number density

Definition at line 93 of file MET_atmosphere_state_vars.hh.

Referenced by METAtmosphereStateVars(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.10.4.5 log10_dens

```
double jeod::METAtmosphereStateVars::log10_dens {}
```

trick_units(–) Log10(total density)

Definition at line 86 of file MET_atmosphere_state_vars.hh.

Referenced by METAtmosphereStateVars(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.10.4.6 mol_weight

```
double jeod::METAtmosphereStateVars::mol_weight {}
```

trick_units(–) Average molecular weight

Definition at line 87 of file MET_atmosphere_state_vars.hh.

Referenced by jeod::METAtmosphere::jacchia(), METAtmosphereStateVars(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.10.4.7 N2

```
double jeod::METAtmosphereStateVars::N2 {}
```

trick_units(–) N2 number density

Definition at line 88 of file MET_atmosphere_state_vars.hh.

Referenced by METAtmosphereStateVars(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.10.4.8 Ox

```
double jeod::METAtmosphereStateVars::Ox {}
```

trick_units(–) O number density

Definition at line 90 of file MET_atmosphere_state_vars.hh.

Referenced by METAtmosphereStateVars(), operator=(), and jeod::METAtmosphere::update_atmosphere().

8.10.4.9 Ox2

```
double jeod::METAtmosphereStateVars::Ox2 {}
```

trick_units(–) O2 number density

Definition at line 89 of file MET_atmosphere_state_vars.hh.

Referenced by METAtmosphereStateVars(), operator=(), and jeod::METAtmosphere::update_atmosphere().

The documentation for this class was generated from the following files:

- [MET_atmosphere_state_vars.hh](#)
- [MET_atmosphere_state_vars.cc](#)

8.11 jeod::METAtmosphereThermal Class Reference

The Thermal aspect of the computation.

```
#include <MET_atmosphere.hh>
```

Public Member Functions

- void [update](#) ()
- double [compute_temperature](#) (double [altitude_km](#))
- [METAtmosphereThermal](#) (const double &[T_exosphere](#), const double &[altitude_km](#))
- virtual [~METAtmosphereThermal](#) ()=default
- [METAtmosphereThermal](#) & [operator=](#) (const [METAtmosphereThermal](#) &)=delete
- [METAtmosphereThermal](#) (const [METAtmosphereThermal](#) &)=delete

Data Fields

- double [T_out](#) {}

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [METAtmosphereThermal](#))
- void [generate_base_temperature](#) ()

Private Attributes

- double [T_125](#) {}
- const double & [T_exosphere](#)
- const double & [altitude_km](#)

Static Private Attributes

- static constexpr double [k_1](#) {0.054285714}
Temperature coefficients.
- static constexpr double [k_3](#) {-3.96501457725948E-5}
- static constexpr double [k_4](#) {-5.3311120366514E-7}
- static constexpr double [T_90](#) {183.0}

8.11.1 Detailed Description

The Thermal aspect of the computation.

Definition at line 131 of file MET_atmosphere.hh.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 METAtmosphereThermal() [1/2]

```
jeod::METAtmosphereThermal::METAtmosphereThermal (
    const double & T_exosphere,
    const double & altitude_km )
```

Definition at line 76 of file MET_atmosphere.cc.

8.11.2.2 ~METAtmosphereThermal()

```
virtual jeod::METAtmosphereThermal::~~METAtmosphereThermal ( ) [virtual], [default]
```

8.11.2.3 METAtmosphereThermal() [2/2]

```
jeod::METAtmosphereThermal::METAtmosphereThermal (
    const METAtmosphereThermal & ) [delete]
```

8.11.3 Member Function Documentation

8.11.3.1 compute_temperature()

```
double jeod::METAtmosphereThermal::compute_temperature (
    double altitude_km )
```

Definition at line 146 of file MET_atmosphere.cc.

References k_1, k_3, k_4, T_125, T_90, and T_exosphere.

Referenced by jeod::METAtmosphere::apply_gauss_quadrature(), jeod::METAtmosphere::jacchia(), and update().

8.11.3.2 generate_base_temperature()

```
void jeod::METAtmosphereThermal::generate_base_temperature ( ) [private]
```

8.11.3.3 operator=()

```
METAtmosphereThermal& jeod::METAtmosphereThermal::operator= (
    const METAtmosphereThermal & ) [delete]
```

8.11.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::METAtmosphereThermal::p (
    jeod ,
    METAtmosphereThermal ) [private]
```

8.11.3.5 update()

```
void jeod::METAtmosphereThermal::update ( )
```

Definition at line 98 of file MET_atmosphere.cc.

References altitude_km, compute_temperature(), T_125, T_exosphere, and T_out.

Referenced by jeod::METAtmosphere::jacchia().

8.11.4 Field Documentation

8.11.4.1 altitude_km

```
const double& jeod::METAtmosphereThermal::altitude_km [private]
```

Definition at line 169 of file MET_atmosphere.hh.

Referenced by update().

8.11.4.2 k_1

```
constexpr double jeod::METAtmosphereThermal::k_1 {0.054285714} [static], [constexpr], [private]
```

Temperature coefficients.

trick_units(1/m) parameter used to obtain the first coefficient of the temperature polynomial, which is also the temperature gradient at 125km.

Definition at line 147 of file MET_atmosphere.hh.

Referenced by compute_temperature().

8.11.4.3 k_3

```
constexpr double jeod::METAtmosphereThermal::k_3 {-3.96501457725948E-5} [static], [constexpr], [private]
```

trick_units(1/m³) parameter used to obtain the 3rd coefficient of the temperature polynomial.

Definition at line 152 of file MET_atmosphere.hh.

Referenced by compute_temperature().

8.11.4.4 k_4

```
constexpr double jeod::METAtmosphereThermal::k_4 {-5.3311120366514E-7} [static], [constexpr],  
[private]
```

trick_units(1/m4) parameter used to obtain the 4th coefficient of the temperature polynomial.

Definition at line 156 of file MET_atmosphere.hh.

Referenced by compute_temperature().

8.11.4.5 T_125

```
double jeod::METAtmosphereThermal::T_125 {} [private]
```

trick_units(K) Temperature at 125km reference point.

Definition at line 163 of file MET_atmosphere.hh.

Referenced by compute_temperature(), and update().

8.11.4.6 T_90

```
constexpr double jeod::METAtmosphereThermal::T_90 {183.0} [static], [constexpr], [private]
```

trick_units(K) Temperature at 90km reference point.

Definition at line 160 of file MET_atmosphere.hh.

Referenced by compute_temperature().

8.11.4.7 T_exosphere

```
const double& jeod::METAtmosphereThermal::T_exosphere [private]
```

Definition at line 166 of file MET_atmosphere.hh.

Referenced by compute_temperature(), and update().

8.11.4.8 T_out

```
double jeod::METAtmosphereThermal::T_out {}
```

Definition at line 134 of file MET_atmosphere.hh.

Referenced by jeod::METAtmosphere::jacchia(), and update().

The documentation for this class was generated from the following files:

- [MET_atmosphere.hh](#)
- [MET_atmosphere.cc](#)

8.12 jeod::WindVelocity::OmegaTableEntry Struct Reference

An entry in an omega scale table.

```
#include <wind_velocity.hh>
```

Data Fields

- double [altitude](#)
Altitude at which omega is multiplied by the corresponding factor.
- double [scale_factor](#)
Factor by which omega is multiplied depending on altitude.

8.12.1 Detailed Description

An entry in an omega scale table.

Definition at line 107 of file wind_velocity.hh.

8.12.2 Field Documentation

8.12.2.1 altitude

```
double jeod::WindVelocity::OmegaTableEntry::altitude
```

Altitude at which omega is multiplied by the corresponding factor.

trick_units(m)

Definition at line 112 of file wind_velocity.hh.

Referenced by jeod::WindVelocity::set_omega_scale_table(), and jeod::WindVelocity::update_wind().

8.12.2.2 scale_factor

```
double jeod::WindVelocity::OmegaTableEntry::scale_factor
```

Factor by which omega is multiplied depending on altitude.

trick_units(-)

Definition at line 117 of file wind_velocity.hh.

Referenced by jeod::WindVelocity::set_omega_scale_table(), and jeod::WindVelocity::update_wind().

The documentation for this struct was generated from the following file:

- [wind_velocity.hh](#)

8.13 jeod::WindVelocity Class Reference

A generic wind velocity implementation.

```
#include <wind_velocity.hh>
```

Data Structures

- struct [OmegaTableEntry](#)
An entry in an omega scale table.

Public Member Functions

- [WindVelocity](#) ()=default
- virtual [~WindVelocity](#) ()
Destructor.
- [WindVelocity](#) (const [WindVelocity](#) &)=delete
- [WindVelocity](#) & operator= (const [WindVelocity](#) &)=delete
- virtual void [update_wind](#) (double inertial_pos[3], double altitude, double wind_inertial[3])
Updates the wind velocity from the parameters given.
- unsigned int [get_num_layers](#) ()
- void [set_omega_scale_table](#) (double altitude, double factor)
- void [set_omega_scale_table](#) (unsigned int [num_layers](#), const double *altitude, const double *factor)
- [OmegaTableEntry](#) * [get_omega_scale_table](#) ()

Data Fields

- bool [active](#) {true}
trick_units(-)
- double [omega](#) {}
The rotational velocity of the planet.

Protected Attributes

- unsigned int `num_layers` {}
Number of altitude layers.
- `OmegaTableEntry` * `omega_scale_table` {}
Table of factors to scale omega based on altitude.

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2` `p` (`jeod`, `WindVelocity`)

Private Attributes

- unsigned int `array_index` {}
last known index into the arrays
- bool `first_pass` {true}
Altitude direction check flag.
- bool `increasing_altitude` {true}
Altitude increasing or decreasing flag.

8.13.1 Detailed Description

A generic wind velocity implementation.

Definition at line 76 of file `wind_velocity.hh`.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 `WindVelocity()` [1/2]

```
jeod::WindVelocity::WindVelocity ( ) [default]
```

8.13.2.2 `~WindVelocity()`

```
jeod::WindVelocity::~~WindVelocity ( ) [virtual]
```

Destructor.

Definition at line 42 of file `wind_velocity.cc`.

References `omega_scale_table`.

8.13.2.3 WindVelocity() [2/2]

```
jeod::WindVelocity::WindVelocity (
    const WindVelocity & ) [delete]
```

8.13.3 Member Function Documentation

8.13.3.1 get_num_layers()

```
unsigned int jeod::WindVelocity::get_num_layers ( )
```

Definition at line 212 of file wind_velocity.cc.

References num_layers.

8.13.3.2 get_omega_scale_table()

```
WindVelocity::OmegaTableEntry * jeod::WindVelocity::get_omega_scale_table ( )
```

Definition at line 248 of file wind_velocity.cc.

References omega_scale_table.

8.13.3.3 operator=()

```
WindVelocity& jeod::WindVelocity::operator= (
    const WindVelocity & ) [delete]
```

8.13.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::WindVelocity::p (
    jeod ,
    WindVelocity ) [private]
```


8.13.3.5 set_omega_scale_table() [1/2]

```
void jeod::WindVelocity::set_omega_scale_table (
    double altitude,
    double factor )
```

Definition at line 217 of file wind_velocity.cc.

References `jeod::WindVelocity::OmegaTableEntry::altitude`, `num_layers`, `omega_scale_table`, and `jeod::WindVelocity::OmegaTableEntry::scale_factor`.

Referenced by `jeod::WindVelocity_wind_velocity_default_data::initialize()`.

8.13.3.6 set_omega_scale_table() [2/2]

```
void jeod::WindVelocity::set_omega_scale_table (
    unsigned int num_layers,
    const double * altitude,
    const double * factor )
```

Definition at line 226 of file wind_velocity.cc.

References `jeod::WindVelocity::OmegaTableEntry::altitude`, `jeod::AtmosphereMessages::framework_error`, `num_layers`, `omega_scale_table`, and `jeod::WindVelocity::OmegaTableEntry::scale_factor`.

8.13.3.7 update_wind()

```
void jeod::WindVelocity::update_wind (
    double inertial_pos[3],
    double altitude,
    double wind_inertial[3] ) [virtual]
```

Updates the wind velocity from the parameters given.

Parameters

in	<i>inertial_pos</i>	The inertial position of the vehicle Units: M
in	<i>altitude</i>	The altitude of the vehicle Units: M
out	<i>wind_inertial</i>	The wind, in the inertial frame, applied to the vehicle Units: M/s

Definition at line 53 of file wind_velocity.cc.

References `active`, `jeod::WindVelocity::OmegaTableEntry::altitude`, `array_index`, `first_pass`, `jeod::AtmosphereMessages::framework_error`, `increasing_altitude`, `num_layers`, `omega`, `omega_scale_table`, and `jeod::WindVelocity::OmegaTableEntry::scale_factor`.

Referenced by `jeod::AtmosphereState::update_wind()`.

8.13.4 Field Documentation

8.13.4.1 active

```
bool jeod::WindVelocity::active {true}
```

trick_units(–)

Definition at line 95 of file wind_velocity.hh.

Referenced by update_wind().

8.13.4.2 array_index

```
unsigned int jeod::WindVelocity::array_index {} [private]
```

last known index into the arrays

Definition at line 137 of file wind_velocity.hh.

Referenced by update_wind().

8.13.4.3 first_pass

```
bool jeod::WindVelocity::first_pass {true} [private]
```

Altitude direction check flag.

trick_units(–)

Definition at line 142 of file wind_velocity.hh.

Referenced by update_wind().

8.13.4.4 increasing_altitude

```
bool jeod::WindVelocity::increasing_altitude {true} [private]
```

Altitude increasing or decreasing flag.

trick_units(–)

Definition at line 147 of file wind_velocity.hh.

Referenced by update_wind().

8.13.4.5 num_layers

```
unsigned int jeod::WindVelocity::num_layers {} [protected]
```

Number of altitude layers.

trick_units(count)

Definition at line 126 of file wind_velocity.hh.

Referenced by get_num_layers(), set_omega_scale_table(), and update_wind().

8.13.4.6 omega

```
double jeod::WindVelocity::omega {}
```

The rotational velocity of the planet.

trick_units(rad/s)

Definition at line 100 of file wind_velocity.hh.

Referenced by jeod::WindVelocity_wind_velocity_default_data::initialize(), and update_wind().

8.13.4.7 omega_scale_table

```
OmegaTableEntry* jeod::WindVelocity::omega_scale_table {} [protected]
```

Table of factors to scale omega based on altitude.

Definition at line 131 of file wind_velocity.hh.

Referenced by get_omega_scale_table(), set_omega_scale_table(), update_wind(), and ~WindVelocity().

The documentation for this class was generated from the following files:

- [wind_velocity.hh](#)
- [wind_velocity.cc](#)

8.14 jeod::WindVelocity_wind_velocity_default_data Class Reference

```
#include <met_data_wind_velocity.hh>
```

Public Member Functions

- [WindVelocity_wind_velocity_default_data](#) ()=default
- void [initialize](#) ([WindVelocity](#) *)
- void [initialize](#) ([WindVelocity](#) &)

Data Fields

- double [omega_scale_fac](#) [[num_layers](#)]
- double [omega_scale_alt](#) [[num_layers](#)]
- double [omega](#) {7.292115146706388e-5}

Static Public Attributes

- static constexpr int [num_layers](#) {12}

8.14.1 Detailed Description

Definition at line 57 of file `met_data_wind_velocity.hh`.

8.14.2 Constructor & Destructor Documentation

8.14.2.1 [WindVelocity_wind_velocity_default_data\(\)](#)

```
jeod::WindVelocity_wind_velocity_default_data::WindVelocity_wind_velocity_default_data ( )
[default]
```

8.14.3 Member Function Documentation

8.14.3.1 [initialize\(\)](#) [1/2]

```
void jeod::WindVelocity_wind_velocity_default_data::initialize (
    WindVelocity & wind_velocity )
```

Definition at line 56 of file `data_met_wind_velocity.cc`.

References `num_layers`, `jeod::WindVelocity::omega`, `omega`, `omega_scale_alt`, `omega_scale_fac`, and `jeod::↔WindVelocity::set_omega_scale_table()`.

8.14.3.2 initialize() [2/2]

```
void jeod::WindVelocity_wind_velocity_default_data::initialize (
    WindVelocity * WindVelocity_ptr )
```

Definition at line 42 of file data_met_wind_velocity.cc.

8.14.4 Field Documentation

8.14.4.1 num_layers

```
constexpr int jeod::WindVelocity_wind_velocity_default_data::num_layers {12} [static], [constexpr]
```

Definition at line 60 of file met_data_wind_velocity.hh.

Referenced by initialize().

8.14.4.2 omega

```
double jeod::WindVelocity_wind_velocity_default_data::omega {7.292115146706388e-5}
```

Definition at line 79 of file met_data_wind_velocity.hh.

Referenced by initialize().

8.14.4.3 omega_scale_alt

```
double jeod::WindVelocity_wind_velocity_default_data::omega_scale_alt[num_layers]
```

Initial value:

```
{180000.0,
    200000.0,
    220000.0,
    240000.0,
    260000.0,
    280000.0,
    300000.0,
    320000.0,
    340000.0,
    360000.0,
    380000.0,
    400000.0}
```

Definition at line 66 of file met_data_wind_velocity.hh.

Referenced by initialize().

8.14.4.4 omega_scale_fac

```
double jeod::WindVelocity_wind_velocity_default_data::omega_scale_fac[num_layers]
```

Initial value:

```
{
    1.0, 1.1, 1.16, 1.21, 1.25, 1.3, 1.34, 1.38, 1.4, 1.405, 1.41, 1.4142136}
```

Definition at line 62 of file met_data_wind_velocity.hh.

Referenced by initialize().

The documentation for this class was generated from the following files:

- [met_data_wind_velocity.hh](#)
- [data_met_wind_velocity.cc](#)

8.15 jeod::WindVelocityBase Class Reference

The generic base class for wind velocity classes.

```
#include <wind_velocity_base.hh>
```

Public Member Functions

- [WindVelocityBase](#) ()=default
- virtual [~WindVelocityBase](#) ()=default
- [WindVelocityBase](#) (const [WindVelocityBase](#) &)=delete
- [WindVelocityBase](#) & operator= (const [WindVelocityBase](#) &)=delete
- virtual void [update_wind](#) (double position[3], double altitude, double wind_inertial[3])

Virtual function to define the interface for inheriting functions.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [WindVelocityBase](#))

8.15.1 Detailed Description

The generic base class for wind velocity classes.

This class has questionable purpose because of its extremely limited capability but is left here for backward compatibility. It should not be used.

Definition at line 77 of file wind_velocity_base.hh.

8.15.2 Constructor & Destructor Documentation

8.15.2.1 WindVelocityBase() [1/2]

```
jeod::WindVelocityBase::WindVelocityBase ( ) [default]
```

8.15.2.2 ~WindVelocityBase()

```
virtual jeod::WindVelocityBase::~~WindVelocityBase ( ) [virtual], [default]
```

8.15.2.3 WindVelocityBase() [2/2]

```
jeod::WindVelocityBase::WindVelocityBase (
    const WindVelocityBase & ) [delete]
```

8.15.3 Member Function Documentation

8.15.3.1 operator=()

```
WindVelocityBase& jeod::WindVelocityBase::operator= (
    const WindVelocityBase & ) [delete]
```

8.15.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::WindVelocityBase::p (
    jeod ,
    WindVelocityBase ) [private]
```

8.15.3.3 update_wind()

```
void jeod::WindVelocityBase::update_wind (
    double position[3],
    double altitude,
    double wind_inertial[3] ) [virtual]
```

Virtual function to define the interface for inheriting functions.

Parameters

in	<i>position</i>	The position of the vehicle, however the specific implementation defines it
in	<i>altitude</i>	The altitude of the vehicle, however the specific implementation defines it
out	<i>wind_inertial</i>	The wind applied to the craft, in the inertial frame

Definition at line 38 of file `wind_velocity_base.cc`.

References `jeod::AtmosphereMessages::framework_warning`.

The documentation for this class was generated from the following files:

- [wind_velocity_base.hh](#)
- [wind_velocity_base.cc](#)

Chapter 9

File Documentation

9.1 atmosphere.hh File Reference

General base class for atmosphere models.

```
#include "environment/time/include/time_standard.hh"
#include "utils/planet_fixed/planet_fixed_posn/include/planet_fixed_posn.↵
hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::Atmosphere](#)
A generic base class for atmospheres.

Namespaces

- [jeod](#)
Namespace jeod.

9.1.1 Detailed Description

General base class for atmosphere models.

9.2 atmosphere_messages.cc File Reference

Implement atmosphere_messages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/atmosphere_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- `#define MAKE\_ATMOSPHERE\_MESSAGE\_CODE(id) JEOD_MAKE_MESSAGE_CODE(AtmosphereMessages, "environment/atmosphere/base_atmos", id)`

9.2.1 Detailed Description

Implement atmosphere_messages.

9.2.2 Macro Definition Documentation

9.2.2.1 MAKE_ATMOSPHERE_MESSAGE_CODE

```
#define MAKE_ATMOSPHERE_MESSAGE_CODE(  
    id ) JEOD_MAKE_MESSAGE_CODE(AtmosphereMessages, "environment/atmosphere/base_atmos", id)
```

Definition at line 35 of file atmosphere_messages.cc.

9.3 atmosphere_messages.hh File Reference

Implement atmosphere_messages.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::AtmosphereMessages](#)

Describes messages used in the [Atmosphere](#) model.

Namespaces

- [jeod](#)

Namespace jeod.

9.3.1 Detailed Description

Implement atmosphere_messages.

9.4 atmosphere_state.cc File Reference

Implementation of the base atmosphere-state model.

```
#include <cstdint>
#include "../include/atmosphere_state.hh"
#include "../include/wind_velocity.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.4.1 Detailed Description

Implementation of the base atmosphere-state model.

9.5 atmosphere_state.hh File Reference

```
#include "environment/time/include/time_standard.hh"
#include "utils/planet_fixed/planet_fixed_posn/include/planet_fixed_posn.↵
hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "atmosphere.hh"
#include "wind_velocity.hh"
```

Data Structures

- class [jeod::AtmosphereState](#)
A generic base class for atmosphere state, containing common atmosphere state parameters, i.e.

Namespaces

- [jeod](#)
Namespace jeod.

9.6 class_declarations.hh File Reference

Forward declarations of classes defined for JEOD 2.0 Atmosphere.

Namespaces

- [jeod](#)

Namespace jeod.

9.6.1 Detailed Description

Forward declarations of classes defined for JEOD 2.0 Atmosphere.

9.7 class_declarations.hh File Reference

Forward declarations of classes defined for JEOD 2.0 Atmosphere.

Namespaces

- [jeod](#)

Namespace jeod.

9.7.1 Detailed Description

Forward declarations of classes defined for JEOD 2.0 Atmosphere.

9.8 data_met_wind_velocity.cc File Reference

```
#include <cstdint>
#include "environment/atmosphere/base_atmos/include/wind_velocity.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "../include/met_data_wind_velocity.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- `#define JEOD\_FRIEND\_CLASS WindVelocity_wind_velocity_default_data`

9.8.1 Macro Definition Documentation

9.8.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS WindVelocity_wind_velocity_default_data
```

Definition at line 21 of file data_met_wind_velocity.cc.

9.9 MET_atmosphere.cc File Reference

Implementation of MET atmosphere model.

```
#include <algorithm>
#include <cmath>
#include <cstdint>
#include <cstring>
#include "environment/time/include/time_utc.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/MET_atmosphere.hh"
#include "environment/atmosphere/base_atmos/include/atmosphere_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.9.1 Detailed Description

Implementation of MET atmosphere model.

9.10 MET_atmosphere.hh File Reference

Implement the MET atmosphere using the atmosphere framework.

```
#include "environment/time/include/time_utc.hh"
#include "utils/math/include/gauss_quadrature.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "MET_atmosphere_state_vars.hh"
#include "environment/atmosphere/base_atmos/include/atmosphere.hh"
```

Data Structures

- class [jeod::METAtmosphereChemical](#)
The chemical composition of the MET [Atmosphere](#).
- class [jeod::METAtmosphereThermal](#)
The Thermal aspect of the computation.
- class [jeod::METAtmosphere](#)

Namespaces

- [jeod](#)

Namespace jeod.

9.10.1 Detailed Description

Implement the MET atmosphere using the atmosphere framework.

9.11 MET_atmosphere_state.cc File Reference

```
#include <cstdint>
#include "utils/message/include/message_handler.hh"
#include "../include/MET_atmosphere_state.hh"
#include "environment/atmosphere/base_atmos/include/atmosphere_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.12 MET_atmosphere_state.hh File Reference

Implement the MET atmosphere state using the atmosphere framework.

```
#include "utils/planet_fixed/planet_fixed_posn/include/planet_fixed_posn.↵
hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "MET_atmosphere.hh"
#include "MET_atmosphere_state_vars.hh"
```

Data Structures

- class [jeod::METAtmosphereState](#)

The MET specific implementation of [AtmosphereState](#).

Namespaces

- [jeod](#)

Namespace jeod.

9.12.1 Detailed Description

Implement the MET atmosphere state using the atmosphere framework.

9.13 MET_atmosphere_state_vars.cc File Reference

Implementation of MET atmosphere model.

```
#include "../include/MET_atmosphere_state_vars.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.13.1 Detailed Description

Implementation of MET atmosphere model.

9.14 MET_atmosphere_state_vars.hh File Reference

Implement the MET atmosphere state variables using the atmosphere framework.

```
#include "utils/planet_fixed/planet_fixed_posn/include/planet_fixed_posn.↵  
hh"  
#include "utils/sim_interface/include/jeod_class.hh"  
#include "environment/atmosphere/base_atmos/include/atmosphere.hh"  
#include "environment/atmosphere/base_atmos/include/atmosphere_state.hh"
```

Data Structures

- class [jeod::METAtmosphereStateVars](#)
The data variables component of the MET specific implementation of [AtmosphereState](#).

Namespaces

- [jeod](#)
Namespace jeod.

9.14.1 Detailed Description

Implement the MET atmosphere state variables using the atmosphere framework.

9.15 met_data_wind_velocity.hh File Reference

```
#include "utils/message/include/message_handler.hh"
```

Data Structures

- class [jeod::WindVelocity_wind_velocity_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.16 solar_max.cc File Reference

```
#include "environment/atmosphere/MET/include/MET_atmosphere.hh"  
#include "../include/solar_max.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) METAtmosphere_solar_max_default_data

9.16.1 Macro Definition Documentation

9.16.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS METAtmosphere_solar_max_default_data
```

Definition at line 23 of file solar_max.cc.

9.17 solar_max.hh File Reference

Data Structures

- class [jeod::METAtmosphere_solar_max_default_data](#)

Namespaces

- [jeod](#)

Namespace jeod.

9.18 solar_mean.cc File Reference

```
#include "environment/atmosphere/MET/include/MET_atmosphere.hh"  
#include "../include/solar_mean.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Macros

- `#define JEOD\_FRIEND\_CLASS METAtmosphere_solar_mean_default_data`

9.18.1 Macro Definition Documentation

9.18.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS METAtmosphere_solar_mean_default_data
```

Definition at line 23 of file solar_mean.cc.

9.19 solar_mean.hh File Reference

Data Structures

- class [jeod::METAtmosphere_solar_mean_default_data](#)

Namespaces

- [jeod](#)

Namespace jeod.

9.20 solar_min.cc File Reference

```
#include "environment/atmosphere/MET/include/MET_atmosphere.hh"  
#include "../include/solar_min.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [JEOD_FRIEND_CLASS](#) METAtmosphere_solar_min_default_data

9.20.1 Macro Definition Documentation

9.20.1.1 JEOD_FRIEND_CLASS

```
#define JEOD_FRIEND_CLASS METAtmosphere_solar_min_default_data
```

Definition at line 23 of file solar_min.cc.

9.21 solar_min.hh File Reference

Data Structures

- class [jeod::METAtmosphere_solar_min_default_data](#)

Namespaces

- [jeod](#)
Namespace jeod.

9.22 wind_velocity.cc File Reference

General base class for wind velocity models.

```
#include <cstdint>  
#include "utils/memory/include/jeod_alloc.hh"  
#include "utils/message/include/message_handler.hh"  
#include "../include/atmosphere_messages.hh"  
#include "../include/wind_velocity.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.22.1 Detailed Description

General base class for wind velocity models.

9.23 wind_velocity.hh File Reference

A wind velocity model based on winds caused by rotation of the planet.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::WindVelocity](#)
A generic wind velocity implementation.
- struct [jeod::WindVelocity::OmegaTableEntry](#)
An entry in an omega scale table.

Namespaces

- [jeod](#)

Namespace jeod.

9.23.1 Detailed Description

A wind velocity model based on winds caused by rotation of the planet.

9.24 wind_velocity_base.cc File Reference

General base class for wind velocity models.

```
#include "../include/wind_velocity_base.hh"  
#include "../include/atmosphere_messages.hh"  
#include "utils/message/include/message_handler.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.24.1 Detailed Description

General base class for wind velocity models.

9.25 wind_velocity_base.hh File Reference

General base class for wind velocity models.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::WindVelocityBase](#)
The generic base class for wind velocity classes.

Namespaces

- [jeod](#)
Namespace jeod.

9.25.1 Detailed Description

General base class for wind velocity models.

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